

- Bacteria have various ways of defending themselves against phage infections, including the CRISPR-Cas system.
- Many animal viruses have an envelope. **Retroviruses** (such as **HIV**) use the enzyme **reverse transcriptase** to copy their RNA genome into DNA, which can be integrated into the host genome as a **provirus**.
- Since viruses can replicate only within cells, they probably evolved after the first cells appeared, perhaps as packaged fragments of cellular nucleic acid.

? Describe enzymes that are not found in most cells but are necessary for the replication of certain types of viruses.

CONCEPT 19.3

Viruses and prions are formidable pathogens in animals and plants (pp. 408–413)

- Symptoms of viral diseases may be caused by direct viral harm to cells or by the body's immune response. **Vaccines** stimulate the immune system to defend the host against specific viruses.
- An **epidemic**, a widespread outbreak of a disease, can become a **pandemic**, a global epidemic.
- Outbreaks of emerging viral diseases in humans are usually not new, but rather are caused by existing viruses that expand their host territory. The H1N1 2009 flu virus was a new combination of pig, human, and avian viral genes that caused a pandemic. The H5N1 avian flu virus has the potential to cause a high-mortality flu pandemic.
- Viruses enter plant cells through damaged cell walls (horizontal transmission) or are inherited from a parent (vertical transmission).
- **Prions** are slow-acting, virtually indestructible infectious proteins that cause brain diseases in mammals.

? What aspect of an RNA virus makes it more likely than a DNA virus to become an emerging virus?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

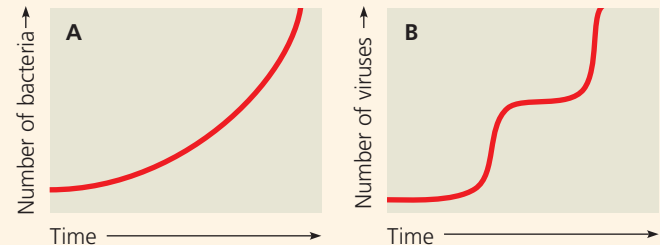
- Which of the following characteristics, structures, or processes is common to both bacteria and viruses?
 - metabolism
 - ribosomes
 - genetic material composed of nucleic acid
 - cell division
- Emerging viruses arise by
 - mutation of existing viruses.
 - the spread of existing viruses to new host species.
 - the spread of existing viruses more widely within their host species.
 - all of the above.
- To cause a human pandemic, the H5N1 avian flu virus would have to
 - spread to primates such as chimpanzees.
 - develop into a virus with a different host range.
 - become capable of human-to-human transmission.
 - become much more pathogenic.

Levels 3-4: Applying/Analyzing

- A bacterium is infected with an experimentally constructed bacteriophage composed of the T2 phage protein coat and T4 phage DNA. The new phages produced would have
 - T2 protein and T4 DNA.
 - T4 protein and T2 DNA.
 - T2 protein and T2 DNA.
 - T4 protein and T4 DNA.
- RNA viruses require their own supply of certain enzymes because
 - host cells rapidly destroy the viruses.
 - host cells lack enzymes that can replicate the viral genome.
 - these enzymes translate viral mRNA into proteins.
 - these enzymes penetrate host cell membranes.
- DRAW IT** Redraw Figure 19.8 to show the replicative cycle of a virus with a single-stranded genome that can function as mRNA (a class IV virus).

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** The success of some viruses lies in their ability to evolve rapidly within the host. Such viruses evade the host's defenses by mutating and producing many altered progeny viruses before the body can mount an attack. Thus, the viruses present late in infection differ from those that initially infected the body. Discuss this as an example of evolution in microcosm. Which viral lineages tend to predominate?
- SCIENTIFIC INQUIRY** When bacteria infect an animal, the number of bacteria in the body increases in an exponential fashion (graph A). After infection by a virulent animal virus with a lytic replicative cycle, there is no evidence of infection for a while. Later, however, the number of viruses rises suddenly and subsequently increases in a series of steps (graph B). Explain the difference in the curves.



- WRITE ABOUT A THEME: ORGANIZATION** While viruses are considered by most scientists to be nonliving, they do show some characteristics of life, including the correlation of structure and function. In a short essay (100–150 words), discuss how the structure of a virus correlates with its function.
- SYNTHESIZE YOUR KNOWLEDGE**



Oseltamivir (Tamiflu), an antiviral drug prescribed for the flu, inhibits the enzyme neuraminidase. Explain how this drug could prevent infection in someone exposed to the flu or could shorten the course of flu in an infected patient (the reasons for which it is prescribed).

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom AAAS
How can genomics help track the Ebola virus?
Go to "Ebola Outbreak Traced to the Funeral of a Traditional Healer"
at www.scienceintheclassroom.org.
➔ **Instructors:** Questions can be assigned in Mastering Biology.